

Sarthak Mishra

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Professional Summary

Computer Science undergraduate with experience building backend systems, real-time monitoring platforms, and full-stack web applications using Python, FastAPI, Flask, SQL, and Linux. Strong foundation in data structures, algorithms, object-oriented programming, and core computer science principles, and an interest in designing reliable, scalable, and maintainable software systems.

Education

Vellore Institute of Technology

Bachelor of Technology in Computer Science and Engineering

CGPA: 7.60/10.00

Sept 2023 – July 2027

Bhopal, India

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks

Technical Skills

Languages: Python, C++, Java, SQL, Bash

Frameworks & Libraries: FastAPI, Flask, Pydantic, Requests, Pytest, Asyncio, Scapy

Databases & Web Technologies: PostgreSQL, SQLite, Redis, REST APIs, HTML, CSS, JavaScript

Developer Tools: Git, GitHub, GitLab, Docker, VS Code, Postman

Operating Systems & Networking: Windows, macOS, Linux, TCP/IP, ARP, DNS, HTTP/HTTPS, SSH

Core Concepts: Data Structures and Algorithms, Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems, Computer Networks

Projects

Hexora – Static File Analysis Platform 🛡️ / Python, FastAPI, SQLAlchemy, SQLite, Docker

- Developed a FastAPI-based web application for static file analysis, capable of analyzing **90+ diverse file samples across multiple formats** through a secure browser-session isolated workflow without executing uploaded content.
- Designed a modular analysis pipeline integrating cryptographic hashing, file signature detection, metadata extraction, entropy analysis, heuristic scoring, and persistent report storage using SQLAlchemy and SQLite.
- Built an interactive web interface supporting detailed analysis reports, searchable history, and containerized deployment using Docker for consistent development and execution.

Arpex – Real-Time Network Monitoring Platform 🛡️ / Python, Flask, Scapy, SQLite, Jinja2

- Developed a real-time network monitoring platform that discovers devices, captures live network traffic, maintains IP-to-MAC mappings, and monitors network state using Python and Scapy.
- Designed a modular backend architecture for device discovery, packet processing, verification workflows, event logging, persistent SQLite storage, and PCAP evidence management.
- Built a Flask-based dashboard to visualize network topology, device inventory, monitoring events, and captured packet evidence through an interactive web interface.

Xentinel – Real-Time System Monitoring Platform 🛡️ / Python, Flask, Chart.js, JSON

- Developed a real-time monitoring platform using Python and Flask to ingest telemetry, process telemetry streams, and generate structured alerts through a modular backend architecture.
- Built an interactive web dashboard with live visualizations, severity-based event monitoring, and JSON-driven analytics using HTML, CSS, JavaScript, and Chart.js.
- Engineered a Linux-based development workflow using Ubuntu, VS Code Remote SSH, and Git to develop, test, and manage monitoring services inside an isolated virtualized environment.

Certifications & Achievements

- Completed the **HackerRank Software Engineer Intern Certificate**.
- Earned **5-Star Gold** badges in both **Python** and **C++** on HackerRank.
- Completed the **Pre Security Learning Path** by TryHackMe.
- Completed **Career Essentials in Software Development** by Microsoft and LinkedIn.

Security Research

- Conducted independent **security research** on **JSTOR's PDF delivery workflow** and privately submitted a responsible disclosure report documenting an authorization workflow observation and remediation recommendations.